

Question Number	Answer	Additional guidance	Mark
6(a)(i)	A description that makes reference to two of the following points: • (at start of interphase) DNA unwrap from histones / {DNA / chromosomes} {uncoil / uncondensed} (1) • DNA {is replicated / synthesis / is checked for errors} (1) • DNA undergoes transcription (1)	ACCEPT genes are {activated / switched on} ACCEPT DNA is methylated / demethylated	(2)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	• metaphase		(1)

Question Number	Answer	Additional guidance	Mark
6(b)(i)	An explanation that makes reference to four of the following points: • centromere doesn't divide (1) • in {anaphase II / meiosis II} (1) • (therefore) both chromosome 13s pulled to same pole of cell / sister chromatids not separated (1) • resulting in {some sperm cells / some nuclei / D} containing two chromosome 13s / {some sperm cells / C} do not have a chromosome 13 (1) • the {egg cell / female nucleus / female gamete} {contains one chromosome 13 / is haploid} (1) • (male and female) nuclei fuse (1)	reject contradictions ACCEPT {some sperm cells / D} have 24 chromosomes ignore genetic material ignore diploid unless specific to just chromosome 13	(4)

Question Number	Answer	Mark
6(b)(ii)	The only correct answer is C totipotent, pluripotent <i>A is not correct because zygote is totipotent</i> <i>B is not correct because blastocyst cells are pluripotent and zygote is totipotent</i> <i>D is not correct because blastocyst cells are pluripotent and zygote is totipotent</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(c)	An answer that makes reference to the following point: error in crossing over has occurred (1)	ACCEPT translocation mutations	(1)

Question Number	Answer	Additional guidance	Mark
6(d)(i)	An explanation that makes reference to the following points: {methyl / -CH₃} groups are added to {gene / cytosine / CpG} (1) - genes become {switched off / deactivated / silenced} (1)	reject histone / lysine / cysteine ACCEPT removal of methyl groups from {gene / cytosine / CpG} ACCEPT gene cannot be {transcribed / expressed} ACCEPT {RNA polymerase / transcription factors} cannot bind ACCEPT heterochromatin formed ACCEPT switching gene on if linked to reduction of methylation	(2)

Question Number	Answer	Additional guidance	Mark
6(d)(ii)	An explanation that makes reference to the following points: - the DNA methylation is copied when the DNA is replicated / {methylated DNA is / epigenetic changes are} replicated (1) (therefore) daughter cells will contain the same DNA methylation after cell division / methylated DNA copies are separated in {anaphase / mitosis} (1)	ACCEPT silenced genes are duplicated ACCEPT daughter cells contain the same {silenced / switched off} genes ACCEPT daughter cells are genetically identical to parent cell ACCEPT mitosis produced genetically identical cells	(2)